LAB 4

C++ Conditional Statements (if / else)

Task 1 | Task 2 | Task 3 | Task 4

Code - Output - Handwritten Copy

Name	**Noor Asma**
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Task 1

Positive / Negative (if-else)

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PROGRAM (SCREENSHOT)

```cpp
#include <iostream>
using namespace std;

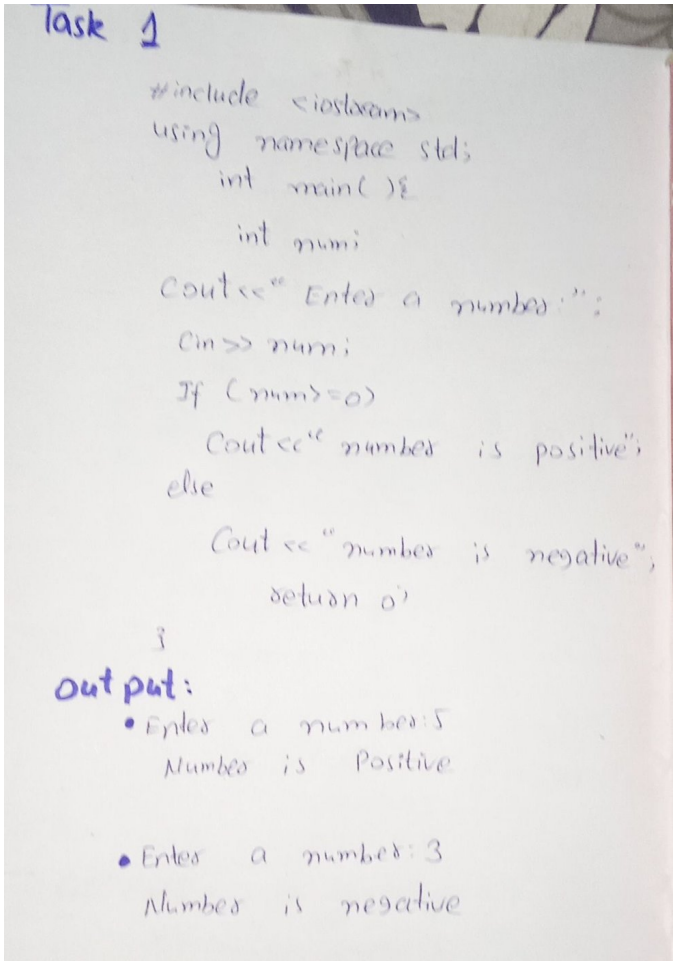
int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num >= 0)
    {
        cout << "Number is positive";
    }
    else
    {
        cout << "Number is negative";
    }

    return 0;
}
```

HANDWRITTEN COPY



OUTPUT (SCREENSHOT)

```
Enter a number:
```

Task 2

Positive / Negative / Zero (else-if ladder)

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PROGRAM (SCREENSHOT)

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter an integer: ";
    cin >> num;

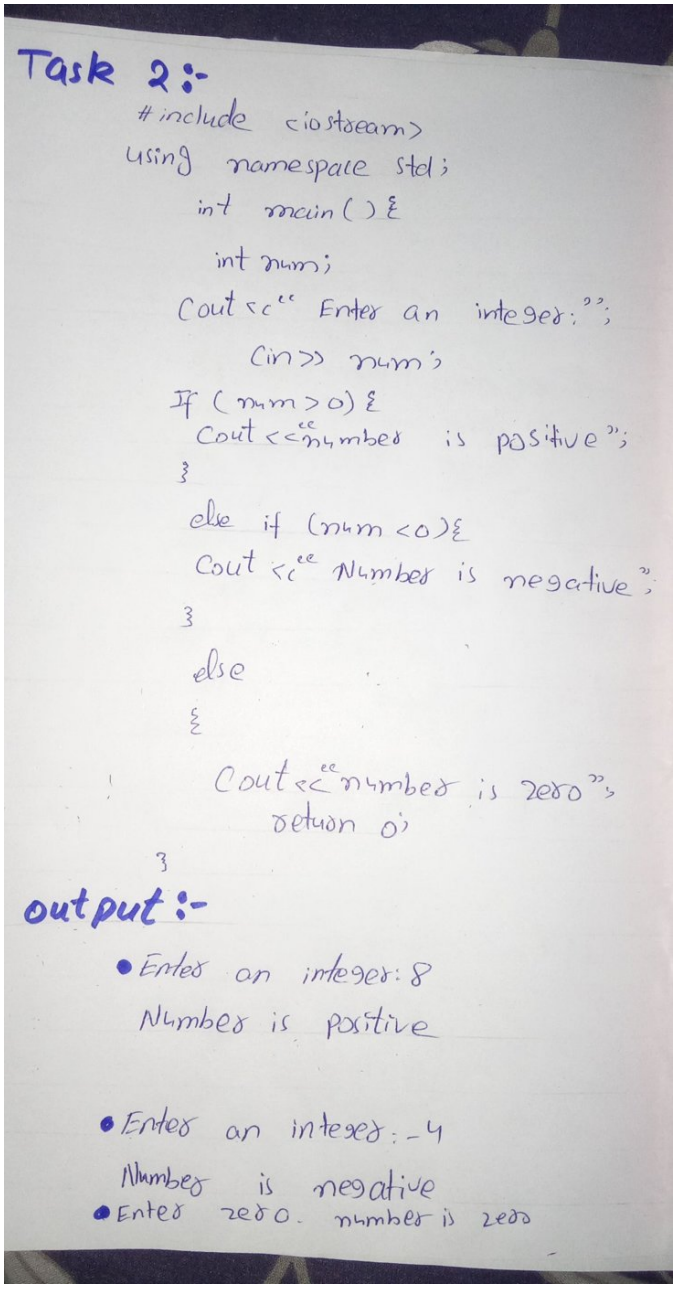
    if (num > 0)
    {
        cout << "Number is positive";
    }
    else if (num < 0)
    {
        cout << "Number is negative";
    }
    else
    {
        cout << "Number is zero";
    }

    return 0;
}
```

OUTPUT (SCREENSHOT)

```
Enter an integer:
```

HANDWRITTEN COPY



Task 3

Grade Calculator (marks)

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PROGRAM (SCREENSHOT)

```cpp
#include <iostream>
using namespace std;

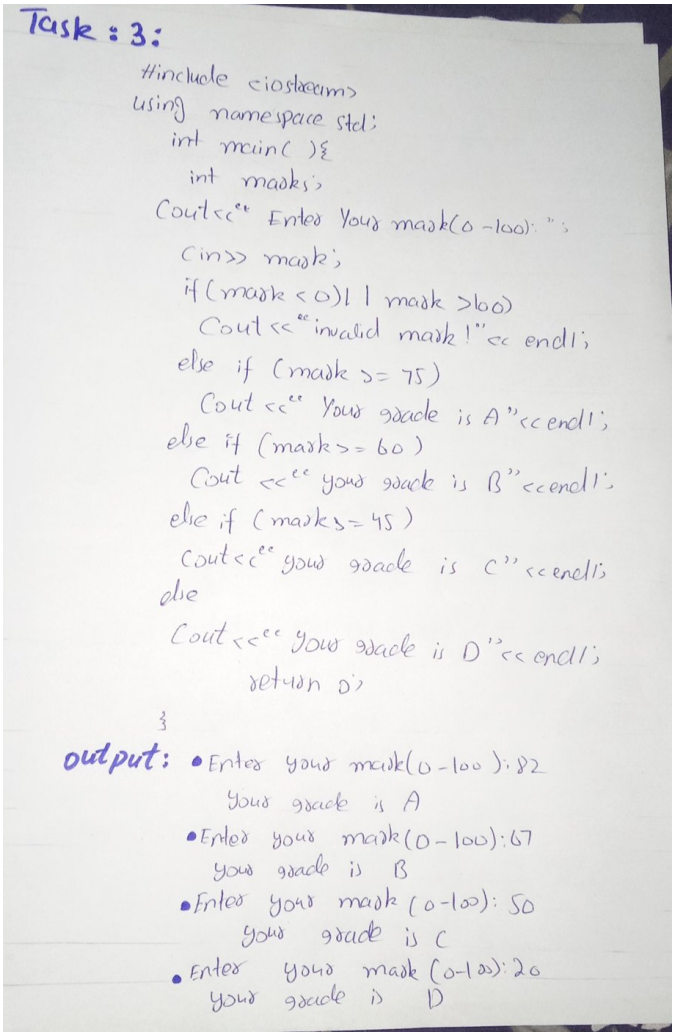
int main()
{
    int marks;

    cout << "Enter your marks: ";
    cin >> marks;

    if (marks >= 75)
    {
        cout << "Your grade is A";
    }
    else if (marks >= 60)
    {
        cout << "Your grade is B";
    }
    else if (marks >= 45)
    {
        cout << "Your grade is C";
    }
    else
    {
        cout << "Your grade is D";
    }

    return 0;
}
```

HANDWRITTEN COPY



Task : 3:

```
#include <iostream>
using namespace std;
int main( ){
  int marks;
Cout<<" Enter Your mark(0-100): ";
 Cin>> mark;
 if(mark < 0)|| mark >100)
  Cout<<" invalid mark!"<< endl;
 else if (mark >= 75)
  Cout<<" Your grade is A"<<endl;
 else if (mark >= 60)
  Cout <<" your grade is B"<<endl;
 else if (mark >= 45)
  Cout<<" your grade is C"<<endl;
 else
  Cout<<" your grade is D"<< endl;
    return 0;
}
```

output:

- Enter your mark(0-100): 82
  your grade is A
- Enter your mark(0-100): 67
  your grade is B
- Enter your mark (0-100): 50
  your grade is C
- Enter your mark (0-100): 20
  your grade is D

OUTPUT (SCREENSHOT)

```
Enter your marks:
```

Task 4

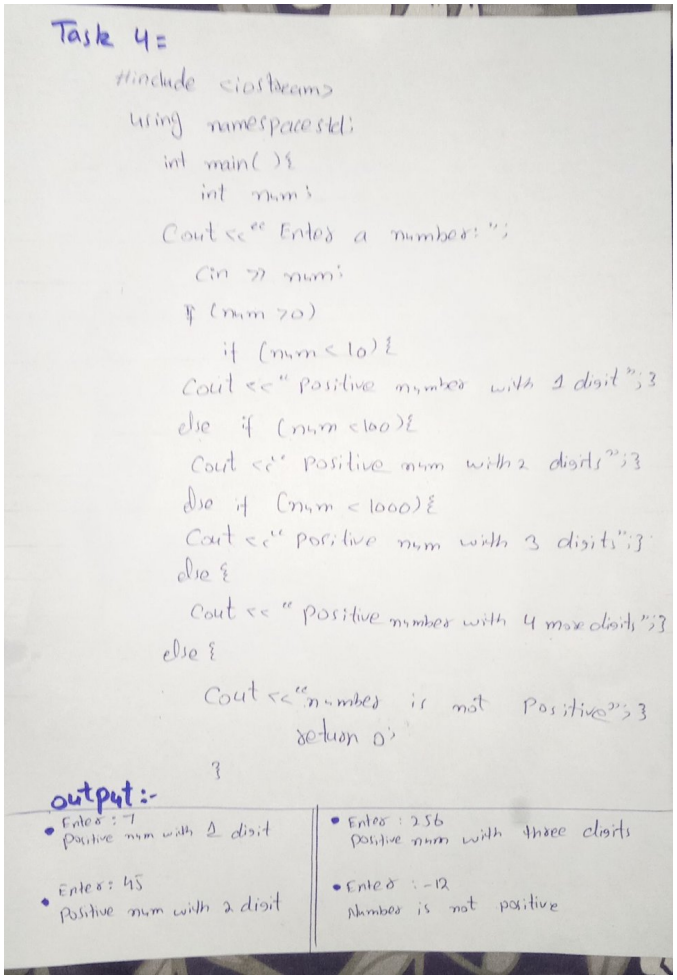
Digit Counter (nested if)

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PROGRAM (SCREENSHOT)

```
4  {
5      int num;
6     cout << "Enter a number: ";
7      cin >> num;
8      if (num > 0)
9      {
10         if (num < 10)
11         {
12             cout << "Positive number with 1
   digit";
13         }
14         else if (num < 100)
15         {
16             cout << "Positive number with 2
   digits";
17         }
18         else if (num < 1000)
19         {
20             cout << "Positive number with 3
   digits";
21         }
22         else
23         {
24             cout << "Positive number with 4 or
   more digits";
25         }
26     }
27     else
28     {
29         cout << "Number is not positive";
30     }
31
32     return 0;
33 }
```

HANDWRITTEN COPY



OUTPUT (SCREENSHOT)

```
Enter a number:
```